G. PULLAIAH COLLEGE OF ENGINEERING AND TECHNOLOGY
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UNIT II

Model Answers for the 10-Mark Question Bank

Derivations · formulas · diagrams · examination keywords

Machine-Learning Basics and Deep Feedforward Networks

Q1 Bias and variance diagnosis from training and validation error

Q2 Validation strategy without contaminating the test set

Q3 Batch gradient descent versus SGD for large networks

Q4 A model fails on unseen data: the first intervention

Q5 Maximum likelihood, output units and losses

Q6 Design of a multiclass feedforward network

Q7 Depth, width, hidden units and representational capacity

Q8 End-to-end training workflow for a fully connected network

Every answer is written to the 10-mark pattern: definition, derivation, formula, diagram, comparison table and conclusion, followed by the keywords that examiners look for.

08
MODEL ANSWERS

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NUMBERED FORMULAE

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VECTOR DIAGRAMS

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The unit at a glance

Q	Core idea	Result to memorise
1	Two error numbers decide everything	gap $= E_{val} - E_{train}$; large gap $\Rightarrow$ variance
2	Parameters, hyperparameters and the estimate need separate data	train $\rightarrow$ parameters, val $\rightarrow$ hyperparameters, test once
3	Noise variance falls as $1/B$, updates grow as m/B	$\mathrm{Var}(\hat{g}) = \sigma^2/B$
4	A low training error rules out the capacity explanation	first fix variance: data, augmentation, regularisation
5	Pick the distribution, then the loss and output unit follow	Gaussian → MSE, Bernoulli → BCE, categorical → CE
6	Softmax plus categorical cross-entropy is forced by the task	$\hat{p} = \mathrm{softmax}(z)$, $J = -\Sigma_k y_k \log \hat{p}_k$
7	Depth buys exponential expressivity, width polynomial	regions $\propto h^L$; parameters linear in L
8	Split, fit on train, then loop forward–loss–backward–update	$\theta \leftarrow \theta - \eta \nabla_\theta J$

HOW TO USE THIS BOOKLET

Each answer follows the same skeleton, so you can revise the *pattern* once and then recall the content: **idea → derivation → formula → diagram → table → conclusion**.

Formula boxes carry equation numbers such as (3.2) — that is *Question 3, second equation*. Figure numbers follow the same rule, so (Fig. 5.1) belongs to Q5.

The **Golden line** is the one sentence to write at the end of an answer; the **Keywords** block is the list an examiner scans for. Aim to reproduce the boxed formulae exactly and to reproduce the diagrams approximately.

Q1 **A model has very low training error but high validation error. Analyse the likely causes using capacity, bias and variance, and propose a systematic diagnosis.**

CO2 · L5 (Design) · 10 Marks

Reading the two numbers

Two measurements decide everything: the **training error** E_{train} and the **validation error** E_{val} on data the model never saw. Their *levels* and their *gap* separate the three possible situations.

$$\text{gap} = E_{val} - E_{train}, \qquad \text{capacity of the hypothesis class } \mathcal{H} \tag{1.1}$$

Observation	Diagnosis	Dominant term
E_{train} high, E_{val} high, small gap	underfitting	bias
E_{train} very low, E_{val} high, large gap	overfitting	variance
E_{train} low, E_{val} low, small gap	good fit	both acceptable

The case in the question is the **second row**: the network has enough capacity to memorise the training set, so it has learned the noise as well as the signal.

Formal decomposition

For a target $y = f(x) + \varepsilon$ with $\mathbb{E}[\varepsilon] = 0$ and $\mathrm{Var}(\varepsilon) = \sigma^2$, the expected squared error of the trained model $\hat{f}$ at a point x decomposes as

$$\mathbb{E}\left[(y - \hat{f}(x))^2\right] = \left(f(x) - \mathbb{E}\hat{f}(x)\right)^2 + \mathrm{Var}\left(\hat{f}(x)\right) + \sigma^2 \tag{1.2}$$

- **Bias** is the error of the average model: how far the family $\mathcal{H}$ is from the truth. It is reduced by *more capacity*.
- **Variance** is the sensitivity of the fitted model to the particular training sample. Models with many parameters, large weights or long training have high variance.
- **Noise** σ^2 is irreducible; no model can remove it, which is why E_{val} can never reach zero.

Because a model with very low E_{train} has, by construction, small bias on the training points, the observed failure must be **variance**: a different training sample would produce a very different network, and the average of those networks would be much better than any single one.

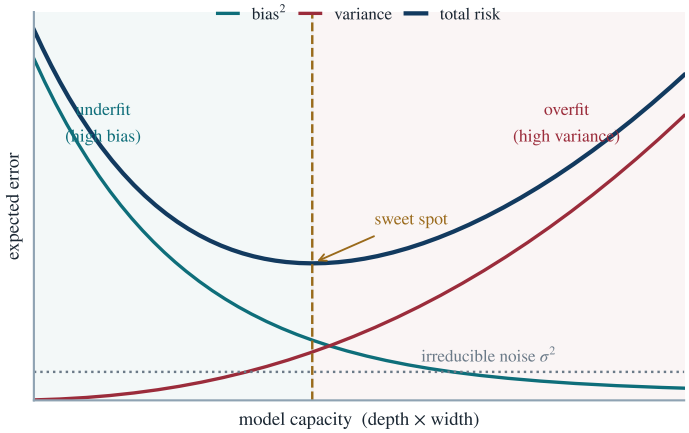


Fig. 1.1 Bias–variance decomposition as a function of capacity. The total risk is U-shaped; overfitting is the right-hand branch.

Why low training error does not imply a good model

1. **Memorisation is cheaper than generalisation.** With p parameters and m samples, a large network can interpolate the training set exactly (even random labels) while learning no transferable structure.
2. **Capacity exceeds data.** The number of parameters is large relative to m, so many parameter settings fit the training data; gradient descent lands in one of them, chosen by the noise it happened to see.

3. **Training error is an optimistically biased estimate.** It is measured on exactly the data used to choose the parameters, so it is the *minimum* of the loss over the sampled set, not its expectation.
4. **Long training with no regularisation** lets weights grow, and large weights amplify input differences — the formal signature of high variance.

Diagnostic procedure

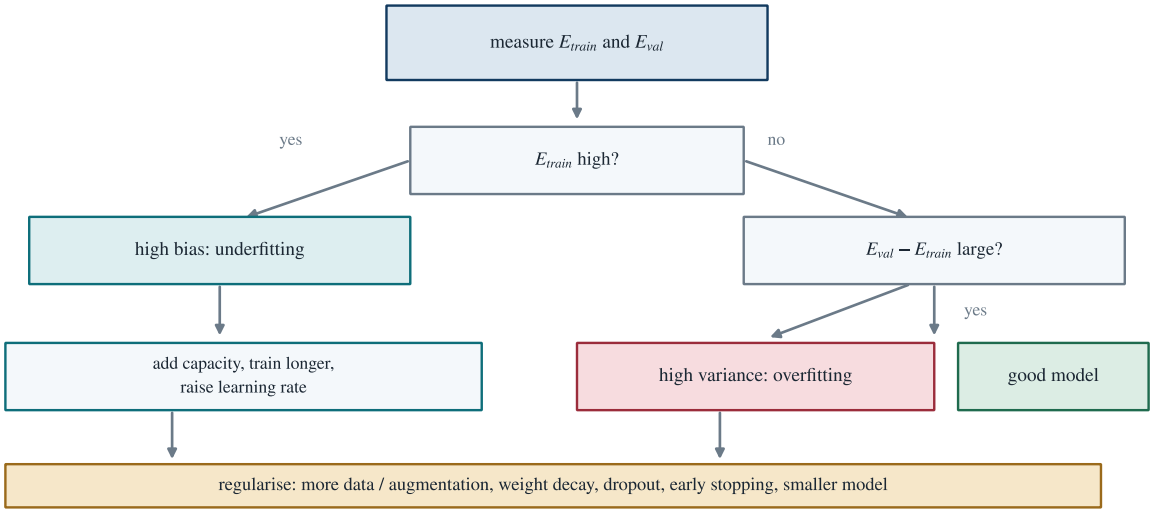


Fig. 1.2 Decision tree from the two error measurements to the appropriate family of interventions.

1. **1 — Verify the measurement.** Recompute both errors with the same preprocessing, the same loss and *model.eval()* (dropout and batch-norm behave differently in training mode). A "validation" error computed with the network in training mode is meaningless.
2. **2 — Check for leakage.** If validation accuracy is suspiciously *high*, features derived from the target, duplicate rows or preprocessing fitted on the whole dataset may be the cause. Confirm that the mean, scaling and PCA used for validation were fitted on the training split only.
3. **3 — Compute the gap explicitly.** A gap of a few percent can be normal; a gap where $E_{val} \approx 2\text{–}3 \times E_{train}$ is a genuine variance problem.
4. **4 — Plot learning curves** of E_{train} and E_{val} against training-set size.

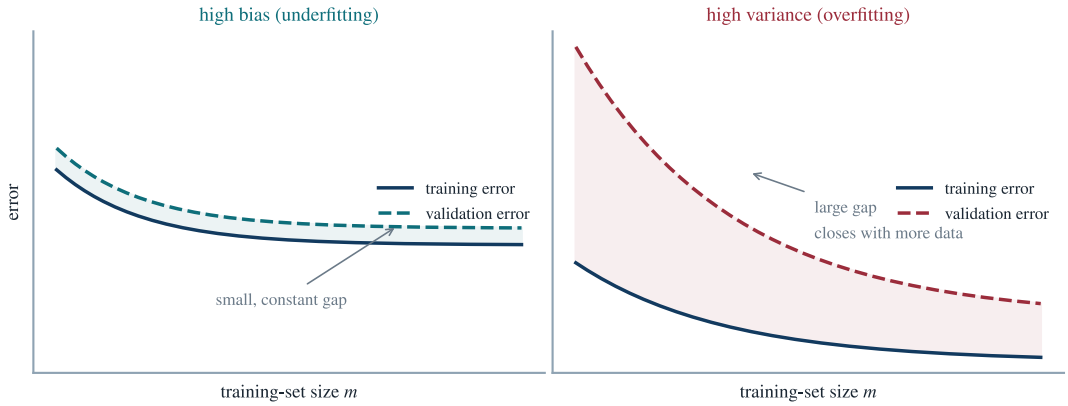


Fig. 1.3 Left: high bias — both curves plateau high with a small gap. Right: high variance — the curves are far apart and the gap closes as data is added.

1. **5 — Test the capacity hypothesis.** Deliberately shrink the model (fewer units/layers) or train with strong weightdecay; if E_{val} improves while E_{train} worsens, variance is confirmed.
2. **6 — Test the data hypothesis.** Train on a subset (e.g.\ 25\%) and on the full set. If the gap narrows when data is added, the diagnosis is "insufficient data for this capacity".
3. **7 — Cross-check with k-fold cross-validation.** A large standard deviation of the validation score across folds is direct evidence of high variance.

Remedies matched to the diagnosis

| Lever | Effect on bias | Effect on variance | Typical setting |

| More training data / augmentation | none | strong decrease | the first intervention when possible |

| Weight decay (ℓ_2) | small increase | decrease | $\lambda \in [10^{-5}, 10^{-2}]$ |

| Dropout | small increase | decrease | 0.2–0.5 on wide layers |

| Early stopping | small increase | decrease | stop at minimum validation loss |

| Smaller architecture | increase | decrease | cut width before depth |

| More capacity | decrease | **increase** | only if E_{train} is also high |

| Longer training | decrease | **increase** | only for underfitting |

Worked example

A $784 \to 512 \to 256 \to 10$ network on 5,000 training images reaches $E_{train} = 0.4\%$ and $E_{val} = 14\%$. The gap of 13.6 percentage points with a near-zero training error is the textbook signature of **high variance**: the model has enough capacity to store the training set.

1. Confirmation: 5-fold cross-validation gives $13.1\% \pm 3.8\%$ — a large spread, again variance.
2. First action: data augmentation and weight decay 10^{-4}; E_{train} rises to 1.8%, E_{val} falls to 8.5%.
3. Second action: add dropout 0.3 after each dense layer and stop early at the best validation epoch: $E_{val} = 6.9\%$.
4. Architecture is *not* reduced, because reducing capacity is only justified once the gap has been reduced by cheaper means.

HOW TO STRUCTURE THE ANSWER

State the two numbers, compute the gap, name the diagnosis (high variance), then give the decomposition formula with all three terms, then the diagnostic procedure, then the matched remedies. Naming the wrong direction ("add capacity") is the commonest way to lose the analytical marks.

GOLDEN LINE

Low training error with high validation error means high variance, not high bias. Diagnose with the gap and the learning curves, then reduce capacity via data, regularisation and early stopping — never by adding more parameters.

KEYWORDS TO WRITE IN THE EXAM

Bias · variance · irreducible noise · bias–variance decomposition · capacity · generalisation gap · learning curve · cross-validation · overfitting · underfitting · weight decay · early stopping

Q2 Design a validation strategy for selecting learning rate, hidden-layer size and number of layers without contaminating the test set. Justify every step.

CO2 · L5 (Design) · 10 Marks

Principle

Three different questions need three different data sources. Mixing them is the single most common experimental error.

$$\text{parameters} \leftarrow \text{training data}, \qquad \text{hyperparameters} \leftarrow \text{validation data}, \qquad \text{performance} \leftarrow \text{test data (once)} \tag{2.1}$$

A **parameter** (weight, bias) is learned by the optimiser. A **hyperparameter** (learning rate η, width h, depth L, regularisation λ, batch size) is chosen by the engineer and cannot be learned by gradient descent on the training loss, because the training loss always prefers more capacity.

Step 1 — Lock the test set away

Split the data before anything else, and do not compute a single statistic on the test part until the very end.

$$\mathcal{D} \;\rightarrow\; \mathcal{D}_{train} + \mathcal{D}_{test} \quad \text{(80/20 or 90/10)} \tag{2.2}$$

Justification: every look at test performance is information, and any decision made using that information makes the test set part of model selection. It then no longer measures generalisation; it measures how well the model was fitted to the test set, which is optimistically biased. The bias grows with the number of models compared — selecting the best of 50 models on the test set is a 50-way fit.

Step 2 — Split the training pool into train and validation

$$\mathcal{D}_{train} \;\rightarrow\; \mathcal{D}_{fit} + \mathcal{D}_{val} \quad \text{(80/20)} \tag{2.3}$$

Parameters are fitted on $\mathcal{D}_{fit}$; hyperparameter configurations are ranked on $\mathcal{D}_{val}$.

Justification: $\mathcal{D}_{val}$ plays the role of unseen data during development, so a decision taken on it stays honest for the final test evaluation. Using a separate validation set from a *different source* (e.g.\ another year or site) is even better, because it also exposes distribution shift.

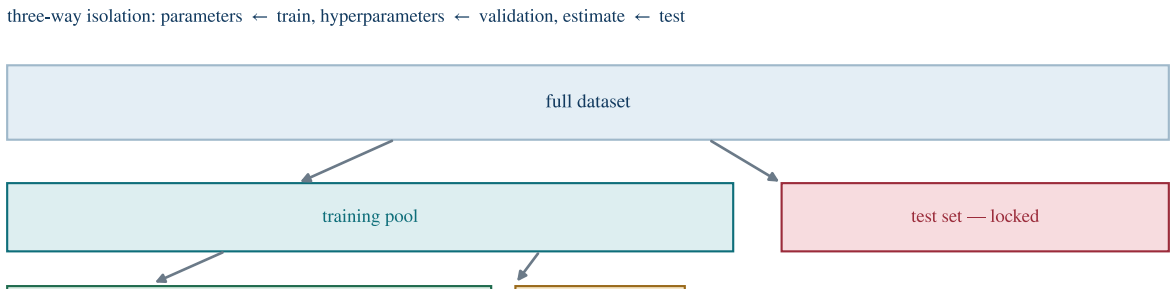


Fig. 2.1 Three-way isolation of the data: parameters from the training pool, hyperparameters from validation, final estimate from the untouched test set.

Step 3 — Choose the search space in a justified way

Hyperparameter	Candidate grid	Justification
Learning rate η	$\{10^{-1}, 3\times10^{-2}, 10^{-2}, 3\times10^{-3}, 10^{-3}, 10^{-4}\}$	logarithmic: the useful range spans orders of magnitude (Question 5, Unit I)
Hidden size h	$\{32, 64, 128, 256, 512\}$	powers of two, again logarithmic in effect
Depth L	$\{1, 2, 3, 4\}$	small integers; depth beyond 4 rarely helps on tabular data
Batch size B	$\{32, 64, 128\}$	hardware efficiency versus gradient noise (Question 3)
Regularisation λ	$\{0, 10^{-5}, 10^{-4}, 10^{-3}\}$	log scale; only after the three main ones are fixed

Justification for the ordering: the learning rate dominates; a model with a bad η looks bad whatever its size. Therefore tune η first with a modest architecture, then width, then depth, then regularisation — a coordinate search instead of a full grid. A full grid costs

$$N_{configs} = |\eta| \times |h| \times |L| = 6 \times 5 \times 4 = 120 \text{ training runs} \tag{2.4}$$

which is why the coordinate order matters in practice.

Step 4 — Use k-fold cross-validation when data is scarce

If the training pool is small, a single 80/20 split wastes data and gives a noisy estimate. Use $k = 5$ (or 10) folds:

$$\mathrm{CV}(\theta) = \frac{1}{k}\sum_{i=1}^{k} \mathrm{Err}\left(f^{(-\mathcal{F}_i)}, \mathcal{F}_i\right) \tag{2.5}$$

Each configuration is trained k times, each time holding out a different fold; the average and standard deviation over folds give a mean and an uncertainty for the configuration.

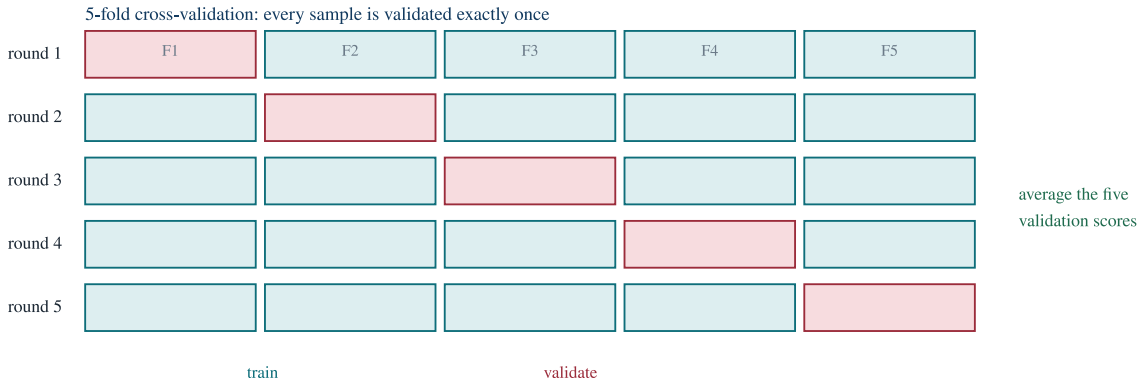


Fig. 2.2 k-fold cross-validation: every sample is used for training $k-1$ times and for validation exactly once.

Justification: all samples contribute to validation exactly once, so the estimate uses the whole pool, and the fold-to-fold standard deviation quantifies how sensitive the choice is. The cost is a factor k in compute.

Step 5 — Fit every preprocessing step inside the fold

The mean, the standard deviation, the PCA basis and any imputation must be fitted on the training part of each fold and merely *applied* to the validation part.

$$\mu_{fold} = \mathrm{mean}(\mathcal{D}_{fit}^{(fold)}), \qquad z = \frac{x - \mu_{fold}}{\sigma_{fold}} \tag{2.6}$$

Justification: computing the scaler on the full dataset leaks the validation distribution (its mean and variance) into training, which produces an optimistic estimate — a subtle but real form of contamination, even though the labels are never seen.

Step 6 — Select, then freeze, then retrain

1. Rank all configurations by mean validation score; select $\theta^{*} = \arg\min_{\theta} \mathrm{CV}(\theta)$.
2. **Freeze** η, h, L and λ. No further changes may be driven by any later measurement.
3. Optionally retrain the selected configuration on the whole training pool ($\mathcal{D}_{fit} \cup \mathcal{D}_{val}$) for the same number of epochs found best — this uses the extra 20\% of data. The retrained model is the final one.
4. With **nested cross-validation**, the inner loop selects hyperparameters and the outer loop estimates performance, which removes the optimism of having selected on the same folds.

Step 7 — Evaluate once on the test set

$$\text{Test error} = \mathrm{Err}(f_{\theta^{*}}, \mathcal{D}_{test}) \tag{2.7}$$

Report it with its confidence interval, and report the validation score alongside. If the test score is much worse than the validation score, the likely cause is a distribution difference between the splits, not a bug in the model.

Strategy	Data used for validation	Bias of the estimate	Variance of the estimate	When to use
Hold-out 80/20	20\% only	moderate (small val set)	high	large datasets, quick iteration
k-fold ($k = 5$)	all data, once each	low	moderate	standard choice
Leave-one-out	$m - 1$ per fold	lowest	high, very expensive	tiny datasets
Stratified k-fold	class-balanced folds	low	lower	imbalanced classes
Time-series split	past only	low	moderate	temporal data (no future leakage)
Nested CV	inner selects, outer estimates	lowest	moderate | expensive	small data + many hyperparameters

CONTAMINATION CHECKLIST

(i) Never tune on the test set. (ii) Never fit scalers/PCA on validation or test data. (iii) Never use test performance to decide when to stop training. (iv) Never report the best of several test evaluations as "the" result. (v) Never shuffle time-series data before splitting.

GOLDEN LINE

Parameters learn from training data, hyperparameters are chosen on validation data, and the test set is opened exactly once. Every preprocessing statistic must be fitted inside the training fold to avoid silent leakage.

KEYWORDS TO WRITE IN THE EXAM

Validation set · hyperparameter · test-set contamination · hold-out split · k-fold cross-validation · stratification · nested CV · data leakage · model selection · generalisation estimate

Q3 Compare batch gradient descent and stochastic gradient descent for training a large deep network. Evaluate their computational behaviour, gradient noise and practical suitability.

CO2 · L5 (Design) · 10 Marks

The two estimators of the same gradient

The loss is an average over the m training samples:

$$J(\theta) = \frac{1}{m}\sum_{i=1}^{m} \ell\big(f(x^{(i)};\theta), y^{(i)}\big) \tag{3.1}$$

Both algorithms perform the same update rule $\theta \leftarrow \theta - \eta\hat{g}$; they differ only in how the gradient estimate $\hat{g}$ is formed.

$$\text{batch: } \hat{g} = \frac{1}{m}\sum_{i=1}^{m}\nabla\ell_i \qquad \text{SGD: } \hat{g} = \nabla\ell_i,\ \ i \sim \text{Uniform}\{1,\dots,m\} \tag{3.2}$$

A mini-batch of size B interpolates between them: $\hat{g} = 1/B\Sigma_{i\in\mathcal{B}}\nabla\ell_i$.

Unbiasedness and noise

Both estimators are unbiased, because the expectation of a sample average is the average:

$$\mathbb{E}[\hat{g}_B] = \nabla J(\theta) \quad \forall B \tag{3.3}$$

The variance of the estimate, however, depends on the batch size:

$$\mathrm{Var}(\hat{g}_B) = \frac{\sigma^2}{B}, \qquad \sigma^2 = \mathrm{Var}(\nabla \ell_i) \tag{3.4}$$

So SGD ($B = 1$) has the largest gradient noise, batch GD ($B = m$) has none, and the standard deviation shrinks as $1/\sqrt{B}$. Halving the noise costs four times the batch size — that square root is the key economic fact.

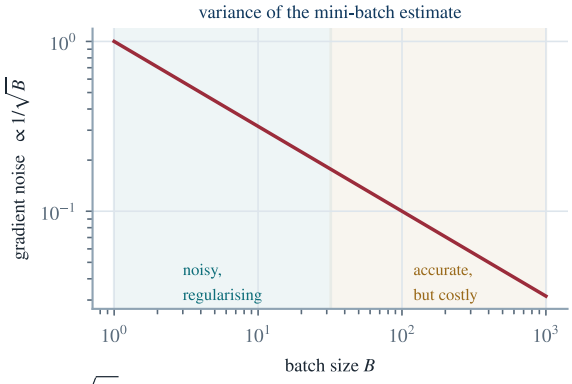


Fig. 3.1 Gradient noise versus batch size: the $1/\sqrt{B}$ law. Small batches are noisy but cheap; large batches are accurate but expensive per step.

Computational behaviour

Aspect	Batch gradient descent	Stochastic / mini-batch GD
Gradient per step	exact, uses all m samples	one sample (or B), noisy
Cost per step	$O(m)$ forward + backward passes	$O(B) \ll O(m)$
Updates per epoch	1	m/B (e.g.\ 500 for $m = 50{,}000$, $B = 100$)
Memory	must hold all activations for all m	only B samples at a time
Hardware use	good vectorisation, but often too large for GPU memory	ideal for GPU/SIMD parallelism
Convergence (smooth convex)	$O(1/k)$, monotone decrease	$O(1/\sqrt{k})$ for the last iterate, but $O(1/k)$ for the averaged iterate
Escape from saddle points	slow — noise-free dynamics stall at $\nabla J = 0$	noise helps escape shallow saddles and poor local minima
Wall-clock to a given loss	usually far slower for large m	usually far faster, despite more iterations

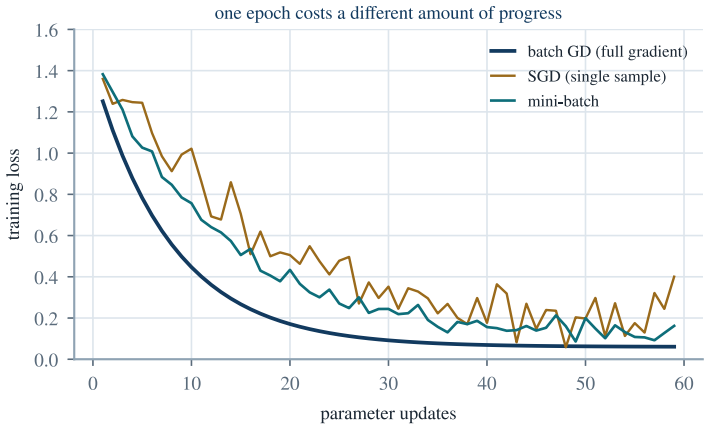


Fig. 3.2 Trace of the training loss against the number of parameter updates. Batch GD moves smoothly; SGD jitters but travels much further per unit of computation.

Why the noise is not purely harmful

1. **Exploration.** At a saddle point $\nabla J = 0$ the batch gradient vanishes and progress stops. The stochastic gradient does not vanish there, so SGD continues and can leave flat regions.

2. **Implicit regularisation.** The noise discourages the solution from settling into a sharp, high-curvature minimum, which tends to generalise better.
3. **Faster early progress.** The loss surface is steep far from the optimum, and a noisy estimate is good enough; the expensive exact gradient buys accuracy where it is not yet needed.

The convergence trade-off, stated properly

For an L-smooth convex objective, batch GD with $\eta = 1/L$ satisfies

$$J(\theta_k) - J^* \le \frac{L\|\theta_0 - \theta^*\|^2}{2k} = O\left(\frac{1}{k}\right) \tag{3.5}$$

whereas SGD with a decaying step $\eta_k \propto 1/\sqrt{k}$ satisfies only

$$\mathbb{E}[J(\theta_k)] - J^* = O\left(\frac{1}{\sqrt{k}}\right) \tag{3.6}$$

But the comparison is unfair in iterations, not time. One batch step costs m gradient evaluations, so per unit of computation the two are comparable (both $O(1/\varepsilon)$), while SGD typically reaches a useful loss far earlier in wall-clock time on large datasets. The noise floor also matters: with a constant η, SGD converges only to a neighbourhood of radius $O(\eta\sigma^2)$, which is why η must decay.

Practical design rules

1. **Batch size.** Use the largest B that fits memory and still gives enough updates per epoch; $B = 32$–256 is the usual range. Bigger batches need a proportionally larger η (**linear scaling rule**: $\eta \propto B$), with a warm-up for the first few hundred steps.
2. **Epochs.** One epoch $= \lceil m/B \rceil$ updates. Report progress in epochs, not iterations, so runs are comparable.
3. **Shuffling.** Reshuffle every epoch; without it the sample order becomes an accidental curriculum and the gradient is biased in blocks.
4. **Learning-rate schedule.** Start moderately, decay (step, cosine or $1/t$). Momentum or Adam stabilises the noisy direction.
5. **Large-batch training.** Very large batches reduce noise so much that generalisation can degrade; remedies are longer warm-up, higher η, or training longer.

Evaluation and conclusion

For a large deep network the *mathematical* objective is the same, but the *engineering* verdict is clear:

- **Batch GD is impractical** beyond small datasets: one update per epoch wastes the epoch, and the memory needed for all activations and the exact gradient is usually unavailable.
- **Pure SGD** ($B = 1$) is too noisy and wastes vectorised hardware.
- **Mini-batch SGD is the practical optimum**: it approximates the full gradient well enough ($\mathrm{Var} \propto 1/B$), exploits GPU parallelism, needs bounded memory, provides m/B updates per epoch, and its noise acts as a useful regulariser and saddle-escape mechanism.

> **MAKE THE EVALUATION EXPLICIT**
> The examiner wants: same update rule, different gradient estimator; unbiasedness; $\mathrm{Var} \propto 1/B$; cost per step $O(B)$ versus $O(m)$; m/B updates per epoch; and the wall-clock verdict. Quote the $1/\sqrt{B}$ noise law — it is the one formula that shows real understanding.

> **GOLDEN LINE**
> Batch GD uses the exact gradient once per epoch; SGD uses a noisy estimate many times per epoch. The noise variance falls as $1/B$, and for large networks mini-batch SGD wins on computation, memory and generalisation.

KEYWORDS TO WRITE IN THE EXAM
Batch gradient descent · stochastic gradient descent · mini-batch · unbiased estimator · gradient noise $\propto 1/\sqrt{B}$ · updates per epoch · convergence rate · saddle point · linear scaling rule · learning-rate schedule

Q4 **A network learns the training examples but fails on unseen examples. Evaluate whether changing the architecture, collecting more data, or changing the learning algorithm is the most appropriate first intervention, and justify your choice.**

CO2 · L5 (Design) · 10 Marks

What the symptom already tells us

"Learns the training examples but fails on unseen examples" is the definition of **overfitting**: $E_{train} \ll E_{val}$ with a large gap (Question 1). The model has sufficient capacity — the training error proves it — so the deficiency is not representational but statistical: the network has fitted sampling noise that does not recur in new data.

$$E_{train} \approx 0, \qquad E_{val} \gg E_{train} \quad \Longrightarrow \quad \text{high variance, low bias} \tag{4.1}$$

That single inference already rules out one of the three options: **changing the architecture is the wrong first move**, because architecture changes are aimed at *capacity*, and capacity is not the missing ingredient.

Step 0 — confirm it is really overfitting

Before intervening, eliminate the three impostors that imitate the symptom:

- **Leakage.** If validation features were built using target information, a "poor" validation score may actually be a broken protocol, not overfitting.
- **Distribution shift.** If validation data come from a different source, the model may be correct but misapplied; the fix is then data or domain adaptation, not regularisation.
- **Label noise / duplicates.** Duplicated rows across splits inflate E_{train} optimism; near-duplicate validation rows hide the true gap.

Only if these are excluded does the diagnosis stand.

Evaluating the three candidate interventions

Intervention	What it changes	Effect on E_{train}	Effect on E_{val}	Cost	Verdict for this symptom
More / better data	the statistical problem itself	rises slightly	falls, gap closes	data collection effort	**first choice** — attacks the cause
Architecture change	capacity, inductive bias	falls	usually rises or unchanged	engineering time	wrong direction here
Learning-algorithm change	optimisation path, implicit regularisation	similar	small, unreliable change	tuning time	secondary

1. **Collecting (or augmenting) data** attacks the cause directly. Variance is the sensitivity of the fit to the particular sample; a larger, more varied sample averages that sensitivity out. The learning-curve test makes this quantitative: if E_{val} is still falling steeply as m grows, more data will help.

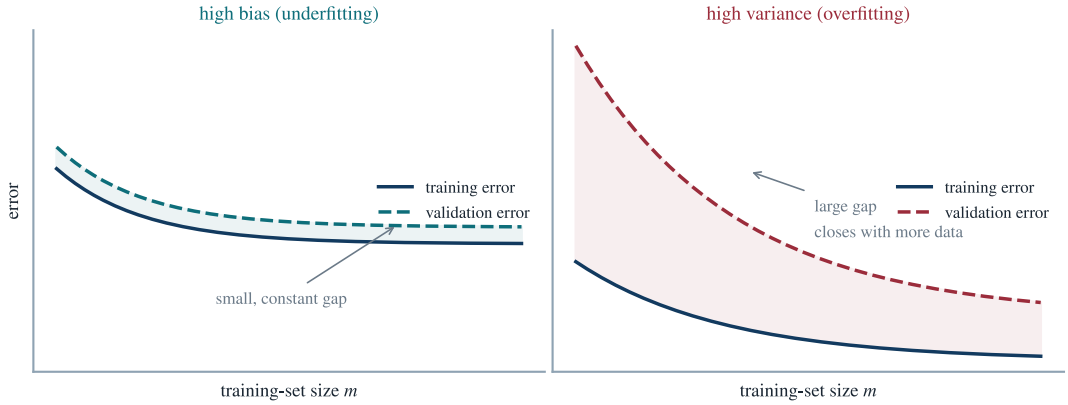


Fig. 4.1 The right-hand panel is this question: a large gap that narrows as training data grows. The left-hand panel would be underfitting, where more data would not help.

1. **Changing the architecture** would be correct only if E_{train} were also high (high bias). A bigger model makes the gap worse; a smaller model can help, but it does so by *reducing capacity*, which is a blunt instrument compared with regularisation and often increases bias unnecessarily.
2. **Changing the learning algorithm** (optimiser, learning rate, initialisation) mostly changes how well the training loss is minimised. Since the training loss is already low, the room for improvement is small; the optimiser is not the bottleneck.

The practical decision order

The three options are not equally cheap, so the justified order is *cheapest effective first*:

1. **1 — Regularise before collecting.** Weight decay, dropout, early stopping and data augmentation are free or nearly so, and they reduce variance immediately. If they close the gap, the data-collection cost is avoided.
2. **2 — Augment before labelling.** Synthetic variation (crops, flips, noise, mixtures) increases effective sample size at almost no cost when it respects the semantics of the task.
3. **3 — Then collect more real data.** The most reliable fix, but the most expensive; justify it by showing that the learning curve is still descending.
4. **4 — Only then reduce capacity** (fewer layers/units), and only if the gap persists.
5. **5 — Changing the learning algorithm** is the last lever, useful mainly when the model is unstable or the loss is not converging at all.

Answer to the "which is most appropriate" question

For a network that already fits the training data:

$$\text{most appropriate first intervention} = \text{increase the effective amount of data} \tag{4.2}$$

(either by **collecting** more examples or, more cheaply, by **augmenting** them and adding **regularisation** which has the same mathematical effect of reducing variance). Changing the architecture is *not* appropriate first, because it targets capacity, and the capacity is demonstrably sufficient; changing the learning algorithm is *not* appropriate first, because optimisation is demonstrably working — the training loss is low.

Justification in bias–variance terms

The regularised objective makes the mechanism explicit:

$$\min_{\theta} \frac{1}{m} \sum_{i=1}^{m} \ell\big(f(x^{(i)};\theta), y^{(i)}\big) + \lambda \|\theta\|^2 \tag{4.3}$$

Increasing m or λ both shrink the effective weight magnitude, which lowers variance at the cost of a small increase in bias. Since this model sits on the low-bias side of the trade-off, paying a little bias is a good trade; adding parameters would move in exactly the wrong direction.

THE TRAP IN THIS QUESTION

Many answers say "the model overfits, so we need a better architecture". That is precisely wrong: a stronger architecture increases capacity and therefore increases variance. The marks are for recognising that the observed training error *rules out* the capacity explanation.

GOLDEN LINE

A low training error with a high validation error is high variance. The first intervention must reduce variance — more data, augmentation and regularisation — not a new architecture (capacity is already sufficient) and not a different optimiser (optimisation is already succeeding).

KEYWORDS TO WRITE IN THE EXAM

Overfitting · generalisation gap · high variance · capacity · data augmentation · regularisation · weight decay · early stopping · learning curve · bias–variance trade-off · distribution shift

Q5 Analyse the relationship among maximum likelihood, output-unit choice and loss-function choice for regression and binary classification in feedforward networks.

CO2 · L4 (Analyse) · 10 Marks

One principle, many losses

Every standard loss function in deep learning is the **negative log-likelihood** of a probabilistic model of the target. Maximum likelihood therefore supplies a single derivation rule that produces the squared error for regression and the cross-entropy for classification, and it simultaneously dictates which output activation must be used.

$$\theta_{ML} = \arg\max_{\theta} \; p(\mathcal{D} \mid \theta) = \arg\max_{\theta} \prod_{i=1}^{m} p\left(y^{(i)} \mid x^{(i)}; \theta\right) \tag{5.1}$$

Taking logarithms (monotone, and numerically safer — Unit I, Question 3) turns the product into a sum:

$$\theta_{ML} = \arg\min_{\theta} \sum_{i=1}^{m} -\log p\left(y^{(i)} \mid x^{(i)}; \theta\right) \tag{5.2}$$

The quantity being minimised **is** the loss. Hence: choose a distribution, and the loss follows automatically. Maximum likelihood is also the estimator that is consistent and asymptotically efficient under correct model specification.

Regression: Gaussian likelihood gives squared error

Assume $y = x^{\top} w + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2)$, so that

$$p(y \mid x; \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y - \hat{y})^2}{2\sigma^2}\right) \tag{5.3}$$

Taking the negative log:

$$-\log p(y \mid x; \theta) = \frac{(y - \hat{y})^2}{2\sigma^2} + \frac{1}{2}\log(2\pi\sigma^2) \tag{5.4}$$

The second term does not depend on θ. Therefore

$$\arg\min_{\theta} \sum_{i} -\log p\left(y^{(i)} \mid x^{(i)}; \theta\right) = \arg\min_{\theta} \sum_{i} \left(y^{(i)} - \hat{y}^{(i)}\right)^2 \tag{5.5}$$

MSE is maximum likelihood under Gaussian noise. The variance σ^2 only scales the loss; the minimiser is unchanged. The **output unit must therefore be the identity** ($\hat{y} = z$), because the Gaussian's mean parameter is unconstrained. Using a

sigmoid on a regression output would restrict $\hat{y}$ to $(0, 1)$ and is a modelling error, not merely a slow start.

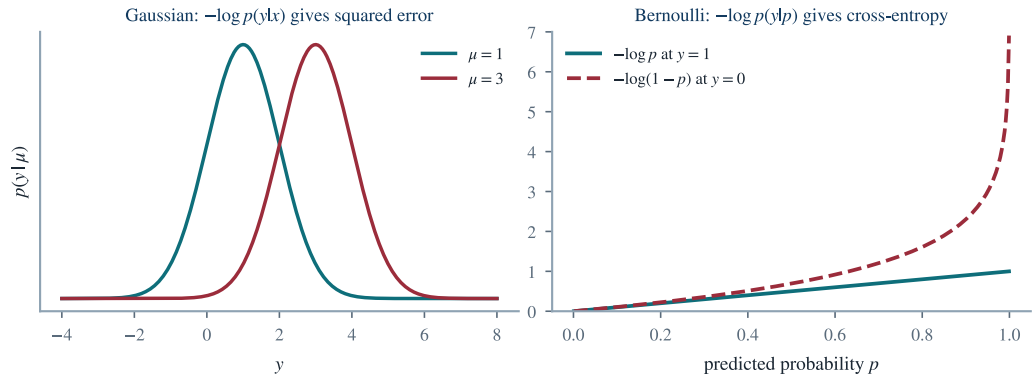


Fig. 5.1 Left: Gaussian likelihood gives the squared-error loss; its mean parameter is unconstrained, so the output unit is linear. Right: Bernoulli likelihood gives the binary cross-entropy; its parameter is a probability, so the output unit is a sigmoid.

Binary classification: Bernoulli likelihood gives cross-entropy

For $y \in \{0, 1\}$ with $p(y = 1 \mid x) = \hat{p} = \sigma(z)$, the Bernoulli likelihood of one example is

$$p(y \mid x) = \hat{p}^{\,y}(1 - \hat{p})^{1-y} \tag{5.6}$$

and its negative log is the **binary cross-entropy**:

$$-\log p(y \mid x) = -[y \log \hat{p} + (1 - y)\log(1 - \hat{p})] \tag{5.7}$$

The output unit must be a sigmoid, because the Bernoulli parameter is a probability in $(0, 1)$ and the sigmoid is exactly the map from the real line to that interval. The two choices are not independent: the *likelihood* fixes the loss, and the *parameter space* fixes the output unit.

Two equivalent ways to see the same loss:

- **Information view:** $\mathrm{CE}(p, q) = H(p) + \mathrm{KL}(p \| q)$; minimising cross-entropy is minimising the KL divergence to the empirical distribution, since $H(p)$ is constant.
- **Gradient view:** with $\hat{p} = \sigma(z)$ and BCE loss, the chain rule collapses beautifully to

$$\frac{\partial L}{\partial z} = \hat{p} - y \tag{5.8}$$

The factor $\sigma'(z) = \hat{p}(1 - \hat{p})$ cancels the $1/\hat{p}$ and $1/(1 - \hat{p})$ of the loss denominators. This is **why sigmoid + BCE must be implemented as a single fused operation**: computing them separately reintroduces the vanishing factor and the numerical instability.

Why the pairing matters (the analysis)

Task	Assumed distribution	Output unit	Loss	Gradient w.r.t. logit
Regression	$\mathcal{N}(\hat{y}, \sigma^2)$	identity	$1/2(y - \hat{y})^2$	$\hat{y} - y$
Binary classification	Bernoulli($\hat{p}$)	sigmoid	binary cross-entropy (Eq. 5.7)	$\hat{p} - y$
Multi-class	Categorical($\hat{p}_k$)	softmax	categorical cross-entropy	$\hat{p}_k - y_k$
Count / Poisson	Poisson(λ)	exponential	$\lambda - y \log \lambda$	$1 - y/\lambda$

The last column is the striking result: **for the matched (canonical) pairing the gradient with respect to the pre-activation is always "prediction minus target".** This is a general property of exponential-family models: the natural (canonical) link makes the Hessian positive semi-definite and the optimisation well conditioned.

What goes wrong when the pairing is mismatched

- **MSE with a sigmoid** (a "matched" loss but the wrong one): the gradient acquires the extra factor $\hat{p}(1-\hat{p})$, so it vanishes exactly when the model is confidently wrong. Learning stalls, and the loss is not convex in the logit.
- **BCE with a linear output** produces log of a value outside $(0, 1)$ — a domain error, or NaN after clamping.
- **Softmax with MSE** on one-hot targets loses the probabilistic interpretation and reduces the effective gradient by an extra factor.
- **Sigmoid + BCE computed separately** overflows/underflows; use *BCEWithLogits* / *log_softmax + NLLLoss* fused forms.

Derivation of the multi-class case (link to Question 6)

With categorical targets $y_k \in \{0, 1\}$, $\Sigma_k y_k = 1$ and $\hat{p}_k = \text{softmax}(z)_k$:

$$-\log p(y \mid x) = -\sum_{k=1}^{K} y_k \log \hat{p}_k \quad \Longrightarrow \quad \frac{\partial L}{\partial z_k} = \hat{p}_k - y_k \tag{5.9}$$

For a one-hot target this is simply $\hat{p}_k$ for the wrong classes and $\hat{p}_k - 1$ for the correct one — an intuitive "push down the wrong, push up the right" update.

Summary of the relationship

1. **Maximum likelihood** is the single principle: minimise the negative log-likelihood of the observed targets.
2. **The assumed distribution** determines the loss: Gaussian → squared error, Bernoulli → binary cross-entropy, categorical → cross-entropy.
3. **The parameter domain of that distribution** determines the output unit: real line → identity, probability → sigmoid, simplex → softmax.
4. Choosing them jointly yields the clean gradient $\hat{y} - y$, convexity in the logits for the exponential-family cases, and numerical stability; choosing them independently breaks one of the three.

ONE SENTENCE THAT CARRIES THE ANSWER

"Maximum likelihood turns a choice of noise model into a choice of loss: Gaussian noise gives squared error with a linear output, Bernoulli noise gives binary cross-entropy with a sigmoid, and categorical noise gives cross-entropy with a softmax — and in each matched case the gradient with respect to the logit is simply prediction minus target."

GOLDEN LINE

Pick the distribution, and the loss and output unit follow from maximum likelihood. Gaussian gives MSE with a linear unit; Bernoulli gives BCE with a sigmoid; categorical gives CE with a softmax.

KEYWORDS TO WRITE IN THE EXAM

Maximum likelihood · negative log-likelihood · Gaussian likelihood · squared error · Bernoulli · binary cross-entropy · softmax · categorical cross-entropy · output unit · canonical link · fused loss · KL divergence

Q6 Design a deep feedforward architecture for multiclass classification and justify the hidden layers, hidden-unit nonlinearity, output layer and objective function.

CO2 · L5 (Design) · 10 Marks

Problem specification

Input $x \in \mathbb{R}^d$, target $y \in \{1, \ldots, K\}$ with $K > 2$ classes and no ordering among them. The design must produce, for every input, a probability vector over the K classes, and it must be trainable by gradient descent.

Architecture

A deep feedforward network with L hidden layers:

$$h^{(1)} = \sigma(W^{(1)}x + b^{(1)}), \quad h^{(l)} = \sigma(W^{(l)}h^{(l-1)} + b^{(l)}), \quad z = W^{(L+1)}h^{(L)} + b^{(L+1)} \tag{6.1}$$

$$\hat{p} = \mathrm{softmax}(z), \qquad \hat{p}_k = \frac{e^{z_k}}{\sum_{j=1}^{K} e^{z_j}}, \qquad \hat{y} = \arg\max_k \hat{p}_k \tag{6.2}$$

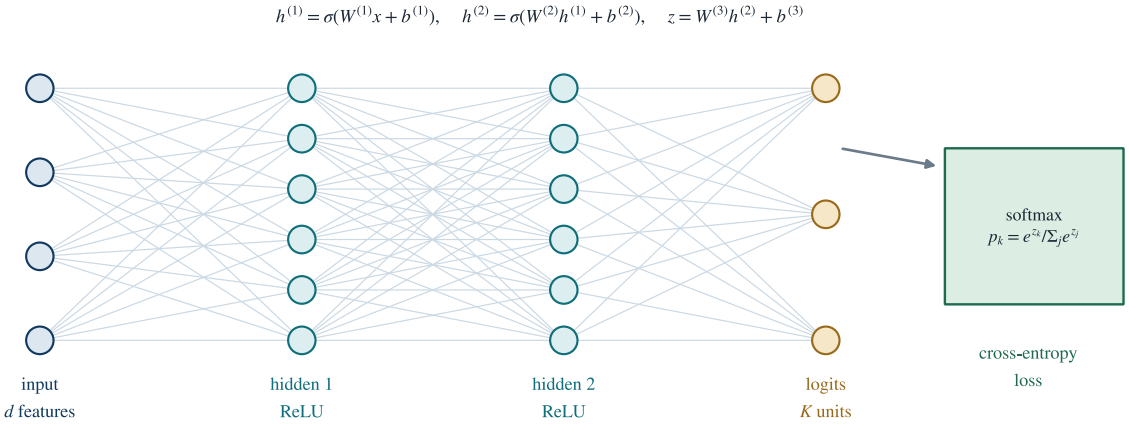


Fig. 6.1 A four-stage design: input, two ReLU hidden layers, a linear layer producing K logits, and a softmax that converts logits into a probability vector.

- ***Parameter count per layer***

$$n_{params}^{(l)} = (n_{l-1} + 1)n_l \tag{6.3}$$

For example $784 \to 256 \to 128 \to 10$ on MNIST: $(784 + 1) \cdot 256 = 200{,}960$, then $257 \cdot 128 = 32{,}896$, then $129 \cdot 10 = 1{,}290$, giving 235,146 parameters in total.

Justification of each design decision

Choice	What was chosen	Justification
Hidden layers (L)	2–3	enough depth to compose nonlinear features; deeper only with residual connections and more data
Hidden width (h)	128–512, decreasing	capacity per layer; decreasing widths compress the representation towards the classifier
Hidden activation	ReLU $\max(0, z)$	non-saturating for $z > 0$, so gradients do not vanish; sparse; cheap; avoids the vanishing-gradient problem of sigmoid/tanh
Output layer	K linear units (logits)	one score per class, unconstrained reals — exactly the domain the categorical model needs
Output activation	softmax	maps $\mathbb{R}^K$ onto the probability simplex: $\hat{p}_k \geq 0$, $\Sigma_k \hat{p}_k = 1$; shift-invariant and stable with max-shift
Objective	categorical cross-entropy	negative log-likelihood of the categorical distribution (Question 5); convex in the logits
Target encoding	one-hot vectors	matches the categorical distribution; avoids implying an order between classes
Prediction rule	$\arg\max_k \hat{p}_k$	Bayes-optimal decision under 0–1 loss

Choice	What was chosen	Justification
Initialisation	He/Xavier	keeps activation variance stable through depth

- ***Why ReLU rather than sigmoid***

$$\sigma'(z) = \sigma(z)(1-\sigma(z)) \le 0.25 \;\Rightarrow\; \text{gradients shrink by } \ge 4\times \text{ per layer} \tag{6.4}$$

ReLU has derivative 1 for $z > 0$ and 0 otherwise, so the gradient of a deep stack does not decay geometrically. Its piecewise-linearity also means the network's output is piecewise linear in the input, which is why deep ReLU networks can represent highly folded decision boundaries.

- ***Why softmax rather than K independent sigmoids***

Sigmoids assume K independent binary decisions. For mutually exclusive classes we require $\Sigma_k p_k = 1$; softmax enforces this exactly and couples the scores, so raising one class's score necessarily lowers the others.

$$\frac{\partial \hat{p}_k}{\partial z_j} = \hat{p}_k\left(\delta_{kj} - \hat{p}_j\right) \tag{6.5}$$

Training objective

$$J(\theta) = -\frac{1}{m}\sum_{i=1}^{m}\sum_{k=1}^{K} y_k^{(i)} \log \hat{p}_k^{(i)} + \frac{\lambda}{2}\|\theta\|^2 \tag{6.6}$$

The first term is the mean negative log-likelihood; the second is optional ℓ_2 regularisation (Unit I, Question 6). Because the loss is the NLL of a categorical distribution, the gradient with respect to the logits is the clean difference

$$\frac{\partial J}{\partial z_k} = \hat{p}_k - y_k \tag{6.7}$$

and back-propagation then continues into $W^{(L+1)}$ and below by the chain rule. In practice the loss is computed as *cross_entropy(logits, targets)*, which fuses log-softmax and NLL to avoid overflow.

Design procedure

1. **Standardise the inputs** and encode labels one-hot; split train/validation/test (Unit II, Question 2).
2. **Fix the output block**: K units + softmax + cross-entropy — this is not a free choice, it follows from the task.
3. **Choose depth and width** from the data size: start shallow ($L = 1$–2, $h \approx \sqrt{d \cdot K}$), then grow while validation error improves (Question 7).
4. **Use ReLU** in hidden layers, He initialisation, and optionally dropout/batch-norm.
5. **Train with mini-batch SGD/Adam**, tune η on validation data, monitor both cross-entropy and accuracy.
6. **Evaluate** with a confusion matrix and per-class precision/recall or macro-F1, not accuracy alone — accuracy hides class imbalance.
7. **Diagnose**: high training loss → underfitting (more capacity); low training loss with high validation loss → overfitting (regularise).

Worked example — design for a 10-class problem with $d = 100$, $m = 20{,}000$

- Architecture: $100 \to 256 \to 128 \to 10 = (101)(256) + (257)(128) + (129)(10) = 59{,}754$ parameters — about 3 parameters per training sample, a reasonable ratio.
- Output: 10 logits, softmax, one-hot targets, categorical cross-entropy.
- Training: Adam, $\eta = 10^{-3}$, batch 128, $\ell_2 = 10^{-4}$, early stopping on validation cross-entropy.
- Expected behaviour: training cross-entropy falls to ≈ 0.05; if validation cross-entropy rises after epoch 12, overfitting has begun and early stopping returns the epoch-12 weights.

DESIGN QUESTIONS WANT REASONS, NOT A LIST

For each block, write *what it is* and *why the task forces it*: softmax because the classes are mutually exclusive; cross-entropy because the likelihood is categorical; ReLU because gradients survive depth; K output units because there are K classes. That "because" is where the marks are.

GOLDEN LINE

Input → ReLU hidden layers → K logits → softmax → categorical cross-entropy, with one-hot targets and arg max prediction. Each choice is forced by the categorical likelihood and by the need for gradients that survive depth.

KEYWORDS TO WRITE IN THE EXAM

Multiclass classification · softmax · logits · one-hot encoding · categorical cross-entropy · ReLU · vanishing gradient · He initialisation · parameter count · confusion matrix · macro-F1

Q7 Analyse how depth, width and hidden-unit choice affect representational capacity in a deep feedforward network.

CO2 · L4 (Analyse) · 10 Marks

Definitions

Capacity is the size and richness of the set of functions $\mathcal{H}$ that the architecture can represent. It is not one number; it is controlled by three knobs:

- **Width** h: number of units per hidden layer.
- **Depth** L: number of hidden layers (parameterised transformations composed).
- **Hidden-unit choice**: which activation and how many units per layer — including non-uniform widths.

$$\text{parameters } P = (d+1)h_1 + \sum_{l=2}^{L}(h_{l-1}+1)h_l + (h_L+1)K \tag{7.1}$$

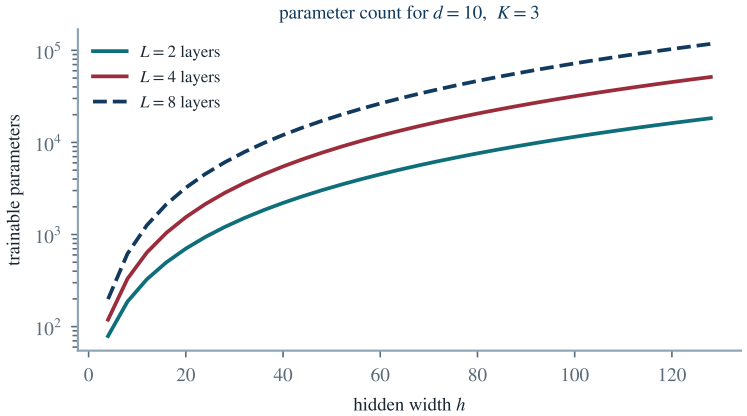


Fig. 7.1 Parameter count against hidden width for three depths ($d = 10$, $K = 3$, logarithmic axis). Every extra layer adds a multiplicative factor at every width, so depth is the cheaper way to grow capacity.

Width and the universal approximation theorem

A network with a single hidden layer of sufficient width and a non-polynomial activation can approximate any continuous function on a compact set to arbitrary accuracy:

$$\forall f \in C(\mathcal{X}),\ \forall \varepsilon > 0\ \ \exists h,\ W:\quad \sup_{x \in \mathcal{X}} \left| f(x) - \sum_{j=1}^{h} w_j \sigma\left(a_j^{\top} x + b_j\right) \right| < \varepsilon \tag{7.2}$$

Interpretation: width is enough, in principle. **But the theorem is not a design recipe** — it says nothing about how many units are needed, whether gradient descent will find them, or how many samples are required. In the worst case the required width grows exponentially with input dimension.

Depth and hierarchical composition

Depth gives something width does not: **composition**. A deep network evaluates

$$f(x) = f_L \circ f_{L-1} \circ \cdots \circ f_1(x) \tag{7.3}$$

so it can build features in stages — edges, then textures, then parts, then objects. The classical result is that some functions representable by a deep network of size poly(d) require exponentially many units in a shallow network. Concretely, a piecewise-linear function with R linear regions needs

$$\text{regions}(L, h) \approx O\left(\left(\frac{h}{h_0}\right)^{L} h^{d}\right) \tag{7.4}$$

so the number of linear pieces — the "folding" of the decision boundary — grows **exponentially with depth** and only polynomially with width. Depth is therefore the more parameter-efficient way to increase capacity.

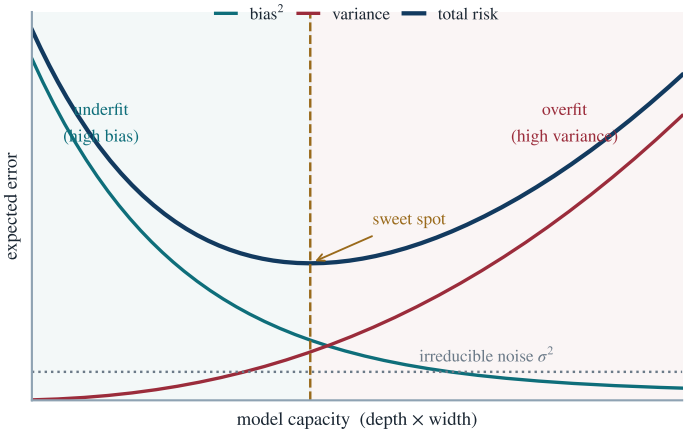


Fig. 7.2 Capacity, whether bought by depth or width, moves the model along the bias–variance trade-off. The optimum is where total risk is minimised, not where capacity is maximised.

The three-way analysis

Property	Increase depth L	Increase width h	Change hidden units / activation
Parameters	grows linearly per added layer	grows quadratically in the layer	depends (wider layer if more units)
Expressivity	exponential gain (composition, linear regions)	polynomial gain	changes smoothness and inductive bias
Feature reuse	strong: later layers reuse earlier features	weak: units act in parallel, not hierarchically	ReLU gives piecewise-linear pieces; tanh gives smooth ones
Optimisation	harder: vanishing/exploding gradients, needs normalisation, residuals, care in initialisation	easier: wider networks are easier to train	saturating units slow learning; ReLU accelerates it
Sample complexity	high — more data needed for the extra flexibility	high	a poor activation can waste capacity entirely
Overfitting risk	high	high	neutral
Practical cost	latency grows with sequential depth	more memory/FLOPs per layer	cheap to change

Effective versus nominal capacity

Nominal capacity (parameter count) is a poor guide on its own. **Effective capacity** is reduced by:

- **Regularisation** — weight decay, dropout, early stopping, augmentation (Unit I, Question 6).
- **Saturation** — with sigmoid/tanh, many units sit in their flat region and contribute little; the usable capacity is far below the nominal one.
- **Dead ReLUs** — units whose pre-activation is negative for all inputs output 0 and can never recover, permanently removing capacity.
- **Data** — capacity is only usable if there are enough samples to estimate the extra parameters. A useful rule of thumb is $P \leq m/10$ for well-conditioned problems, with augmentation relaxing the bound.

Empirically, deep networks generalise even when $P \gg m$ — the "double descent" phenomenon — because the *implicit* regularisation of SGD and the architecture itself restricts which of the many interpolating solutions is found. Capacity theory therefore predicts the *possibility* of overfitting, not its inevitability.

Design guidance

1. **Grow depth before width** when the data has hierarchical structure (images, audio, text), because composition is what such data requires.
2. **Grow width before depth** when the data is tabular and shallow, or when training stability is a concern — wide models are easier to optimise.
3. **Keep the final hidden layer modest** so the classifier head does not dominate the parameter budget; compress progressively.
4. **Match capacity to data**: start small and grow while validation error improves (Question 7 diagnosis), instead of starting large and hoping regularisation saves the model.
5. **Prefer architectural priors over raw capacity**: convolution for images, weight sharing, or residual connections buy expressivity per parameter far more cheaply than adding units.

Worked comparison

Two candidates for a $d = 1{,}000$, $K = 10$ task with $m = 50{,}000$:

- **Wide and shallow:** 1 hidden layer of $2{,}048$ units $\Rightarrow P = (1001)(2048) + (2049)(10) \approx 2.07$ million.
- **Deep and narrow:** 4 hidden layers of 256 units $\Rightarrow P = (1001)(256) + 3(257)(256) + (257)(10) \approx 0.46$ million.

The deep model has $4.5\times$ fewer parameters yet typically represents the task better, because its 4 composed layers create exponentially more linear regions than the single wide layer. The wide model, however, trains more reliably and is preferred when m is small.

THE ANALYSIS THE EXAMINER WANTS

Do not just say "more layers = more capacity". Say *how much more*: linear regions grow as h^L (exponential) but only polynomially in h; parameter count grows linearly in L; and effective capacity is what matters, reduced by regularisation, saturation and dead units.

GOLDEN LINE

Width guarantees approximation; depth guarantees efficiency of approximation. Parameters grow linearly with depth and quadratically with width, while representational power grows exponentially with depth and only polynomially with width.

KEYWORDS TO WRITE IN THE EXAM

Depth · width · hidden units · capacity · universal approximation · hierarchical composition · linear regions · parameter count · vanishing gradient · effective capacity · double descent · overfitting

Q8 **Construct an end-to-end training workflow for a fully connected deep network, starting from dataset partitioning and ending with back-propagation-based parameter updates. Include the key learning decisions that must be fixed before training.**

CO2 · L5 (Design) · 10 Marks

Overview

The workflow has twelve stages. The first six are decisions that must be **fixed before the first gradient step**; the last six form the optimisation loop and its evaluation.

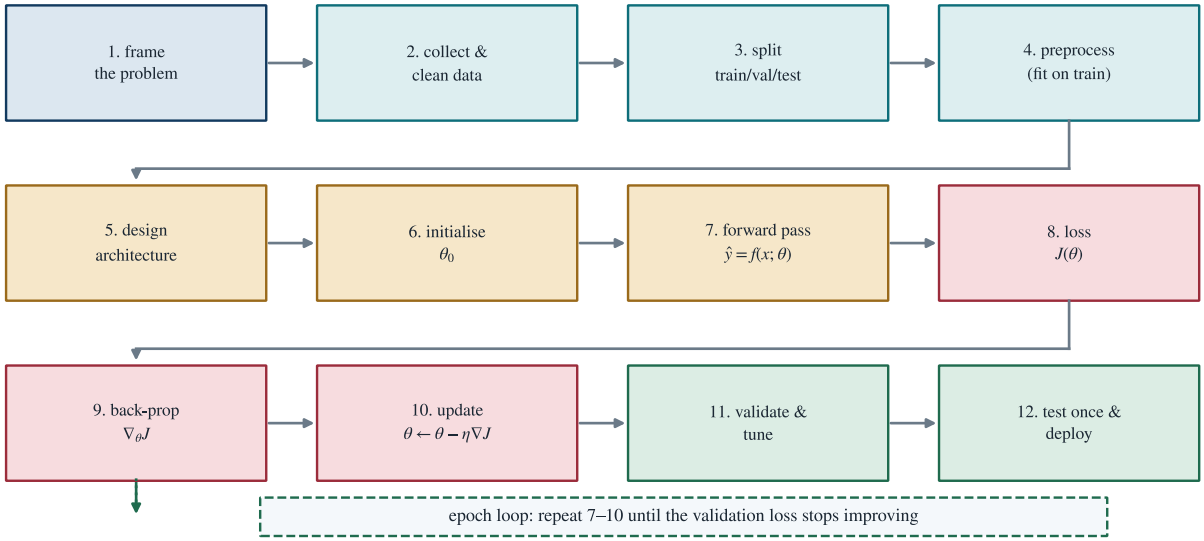


Fig. 8.1 The complete workflow. Stages 1–4 prepare the data, 5–8 define the model and its loss, and 9–12 iterate and evaluate; the dashed arrow is the epoch loop.

Stage 1 — Frame the problem

Fix the input space $\mathcal{X}$, the target space $\mathcal{Y}$ and the loss metric before touching the data. Classification or regression determines the output unit and the loss (Question 5); the metric (accuracy, macro-F1, RMSE) determines what "better" means.

Stage 2 — Collect and clean

Handle missing values, remove duplicates, correct labels, and check class balance. **Duplicates across splits are the most common cause of imaginary accuracy**, so deduplicate before splitting.

Stage 3 — Partition the data

$$\mathcal{D} \rightarrow \mathcal{D}_{train}\ (70\%) + \mathcal{D}_{val}\ (15\%) + \mathcal{D}_{test}\ (15\%) \tag{8.1}$$

Stratify by class, and split *before* any preprocessing so that no statistic crosses the boundary. The test set is opened exactly once, at stage 12.

Stage 4 — Preprocess (fit on training data only)

$$x \leftarrow \frac{x - \mu_{train}}{\sigma_{train}} \tag{8.2}$$

Then optional PCA or feature selection, with every parameter of the transform estimated on $\mathcal{D}_{train}$ and merely applied elsewhere. **Decision to fix now:** whether to standardise, and whether to reduce dimension (Unit I, Question 8).

$$\Sigma_{train} = \frac{1}{m-1} X_{train}^{\top} X_{train}, \qquad Z = XW_k \tag{8.3}$$

Stage 5 — Fix the architecture

Choose depth L, widths h_l, activation σ, and the output layer. Fix the parameter count in advance:

$$P = (d+1)h_1 + \sum_{l=2}^{L}(h_{l-1}+1)h_l + (h_L+1)K \tag{8.4}$$

Decisions to fix: depth, width, activation (ReLU), output unit (softmax or identity), and the parameter budget relative to m (Question 7).

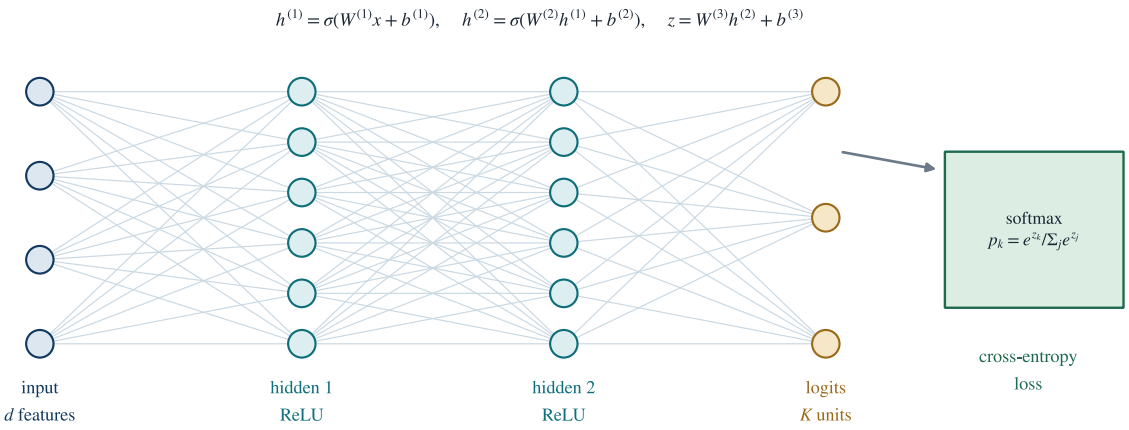


Fig. 8.2 The reference architecture used throughout: ReLU hidden layers, a linear layer producing logits, then softmax and cross-entropy.

Stage 6 — Fix initialisation and regularisation

$$W \sim \mathcal{N}\left(0, \frac{2}{n_{in}}\right) \text{ (He)}, \qquad b = 0 \tag{8.5}$$

Decisions to fix: initialiser (He for ReLU, Xavier for tanh), weight decay λ, dropout rate, and whether to use batch normalisation. These must be chosen before training because they change the optimisation landscape itself.

Stage 7 — Forward pass

For each mini-batch, compute activations layer by layer and cache them (they are needed for back-propagation):

$$a^{(0)} = x, \qquad z^{(l)} = W^{(l)}a^{(l-1)} + b^{(l)}, \qquad a^{(l)} = \sigma(z^{(l)}), \qquad \hat{p} = \text{softmax}(z^{(L+1)}) \tag{8.6}$$

Stage 8 — Compute the loss

$$J(\theta) = -\frac{1}{B}\sum_{i\in\mathcal{B}}\sum_{k=1}^{K} y_k^{(i)} \log \hat{p}_k^{(i)} + \frac{\lambda}{2}\sum_{l}\|W^{(l)}\|^2 \tag{8.7}$$

Decision to fix: the loss, which follows from the likelihood (cross-entropy for classification, MSE for regression), plus the reduction (mean over the batch).

Stage 9 — Back-propagation

Apply the chain rule backwards from the loss to the parameters. The starting point is the clean output-layer error

$$\delta^{(L+1)} = \frac{\partial J}{\partial z^{(L+1)}} = \hat{p} - y \tag{8.8}$$

and then recursively, for $l = L, \ldots, 1$:

$$\delta^{(l)} = (W^{(l+1)\top}\delta^{(l+1)}) \odot \sigma'(z^{(l)}), \qquad \frac{\partial J}{\partial W^{(l)}} = \delta^{(l)}a^{(l-1)\top}, \qquad \frac{\partial J}{\partial b^{(l)}} = \delta^{(l)} \tag{8.9}$$

The cost is one forward and one backward pass per batch — the same order as a single evaluation of the loss, which is what makes training feasible.

Stage 10 — Update the parameters

$$\theta \leftarrow \theta - \eta\,\hat{g}, \qquad \hat{g} = \frac{1}{B}\sum_{i \in \mathcal{B}} \nabla_\theta \ell_i, \qquad \mathrm{Var}(\hat{g}) \propto \frac{1}{B} \tag{8.10}$$

Decisions to fix: optimiser (SGD with momentum or Adam), learning rate η and its schedule, batch size B, number of epochs, and gradient clipping threshold (Question 3).

Stage 11 — Validate and tune

After each epoch, evaluate on $\mathcal{D}_{val}$ in *eval* mode. Track both losses, keep the best checkpoint, and stop when validation loss has not improved for a patience of p epochs. Never tune on $\mathcal{D}_{test}$.

$$\theta^{*} = \arg\min_{\theta}\ \mathrm{Err}_{val}(\theta) \tag{8.11}$$

Stage 12 — Evaluate once and deploy

Evaluate the frozen best checkpoint on $\mathcal{D}_{test}$, report the metric with a confidence interval and a confusion matrix, then monitor the deployed model for distribution drift.

The decisions that must be fixed before training

Decision	Options	Fixed by
Input representation	raw / standardised / PCA-reduced	correlation analysis, validation
Output unit and loss	softmax+CE, sigmoid+BCE, identity+MSE	the target distribution (Q5)
Depth and width	$L \in \{1..4\}$, $h \in \{32..512\}$	validation performance (Q7)
Activation	ReLU / tanh / GELU	depth and stability
Initialisation	He / Xavier	activation function
Regularisation	λ, dropout, early stopping	validation (Q1)
Optimiser and η	SGD+momentum / Adam; η log-grid	validation (Unit I, Q5)
Batch size and epochs	$B \in \{32, 64, 128\}$; $E \in [10, 200]$	memory and learning curve
Data split and metric	70/15/15, stratified; accuracy or F1	task requirements

Complete algorithm

1. Split: $\mathcal{D} \to \mathcal{D}_{train} + \mathcal{D}_{val} + \mathcal{D}_{test}$; deduplicate first.
2. Fit the preprocessing on $\mathcal{D}_{train}$; transform all three splits with those statistics.
3. Initialise θ_0 with He/Xavier; fix η, B, L, h, λ, E.
4. For each epoch $e = 1, \ldots, E$: shuffle $\mathcal{D}_{train}$; for each mini-batch compute stages 7–9 and update by stage 10.
5. After each epoch compute E_{train} and E_{val}; save the weights with the best E_{val}.
6. Stop on patience or on the epoch budget; restore the best checkpoint.
7. Evaluate once on $\mathcal{D}_{test}$; report metric, confusion matrix and the learning curves.

> **WHY THE ORDER MATTERS**
> Every stage constrains the next: the target type fixes the loss, the loss fixes the output unit, the depth fixes the initialisation scheme, and the split fixes what may be measured. Changing a decision after seeing the test score invalidates the estimate — which is why stages 1–6 are called "decisions fixed before training".

> **GOLDEN LINE**
> Split first, fit preprocessing on the training split only, fix architecture/loss/optimiser, then iterate forward pass → loss → back-propagation → update, select on validation, and touch the test set exactly once.

KEYWORDS TO WRITE IN THE EXAM

End-to-end workflow · data partition · preprocessing leakage · architecture · initialisation · forward pass · back-propagation · chain rule · mini-batch · optimiser · early stopping · validation · final test evaluation

Rapid revision — the whole unit in one grid

Read this column-wise the night before the examination: the trigger phrase, the three things that must appear, and the trap that costs marks.

Question	Three things that must appear	The trap
1 Bias–variance	gap $= E_{val} - E_{train}$; the three-term decomposition; learning curves	saying "add capacity" for a model that already fits
2 Validation	parameters/hyperparameters/test separation; k-fold; preprocessing inside the fold	fitting the scaler on all the data
3 Batch GD vs SGD	same update, different estimator; $\mathrm{Var} \propto 1/B$; m/B updates per epoch	claiming SGD converges faster per iteration
4 First intervention	diagnosis from the two errors; variance remedies in cost order; learning-curve test	changing architecture for an overfitted model
5 MLE and losses	NLL principle; Gaussian → MSE; Bernoulli → BCE; output unit follows the parameter	pairing sigmoid with MSE
6 Multiclass network	ReLU hidden layers; K logits; softmax; categorical CE; one-hot targets	independent sigmoids for exclusive classes
7 Depth and width	universal approximation (width); exponential regions (depth); parameter formulas	equating capacity only with parameter count
8 End-to-end workflow	split first; forward/loss/backward/update; validate; test once; the pre-training decisions	tuning on the test set